



Housing Price Prediction — Summary of Findings

Dataset: Housing.csv (545 records) | Target: price | Models: Linear Regression & Random Forest

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.6529
Random Forest	1,014,947	1,399,769	0.6124

Which features influence price the most?

Area (plot size) is the dominant predictor, accounting for roughly 47% of the Random Forest model's predictive power. Number of bathrooms (15%) and air conditioning (6%) follow as the next strongest drivers, with preferred-area location, basement, and parking contributing meaningfully as well.

How accurate was the model?

Linear Regression performed best, with an R² of 0.653 — explaining about 65% of the variation in house prices. Predictions were off by roughly ₹970,000 on average (MAE). Across a price range of ₹1.75M to ₹13.3M, this is a reasonable baseline that captures the broad trend while missing finer nuances.

What was surprising?

Random Forest did not outperform Linear Regression — unusual, and a sign that price relationships in this dataset are largely linear rather than complex or non-linear. The dataset also had zero missing values and zero duplicates, which is uncommon for real-world housing data and suggests it may be curated or synthetic.

Recommendation for a real estate business

Since area is by far the strongest price predictor, agencies should prioritize accurate square-footage data and use it as the primary filter in automated valuation models. Highlighting air conditioning and bathroom count in listings can also help justify premium pricing, as these show the strongest correlation with price after area.